Sinan Koparan

PhD Candidate | Sports Data Science & AI | University of Technology Sydney

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Profile

I am a PhD Candidate in Sports Data Science and AI in the Next Generation Graduates Program (NGGP), supported by CSIRO Data61 and Rugby Australia. My research investigates how data science and machine learning can be used to strengthen community sport participation by identifying how club organisational characteristics influence participation growth and retention. The aim is to help national sporting organisations make evidence-based resource allocation decisions to improve retention, grow participation, and support long-term club sustainability.

Alongside my PhD, I work as a Research Assistant on an applied AI project in partnership with the NSW Institute of Sport (NSWIS), exploring how Large Language Models can be leveraged in coach and athlete environments to automate content analysis and support more efficient insight generation and decision-making.

Research Experience

PhD Candidate / Embedded Research Partner 2024 – present

Rugby Australia Community Rugby · University of Technology Sydney

- Investigating club organisational characteristics and their influence on participation growth and retention in community sport
- Embedded with the Rugby Australia Community Rugby team, translating data insights into actionable strategies
- Developing interactive dashboards and visualisations to monitor participation, retention, and club performance
- Collaborating directly with the national team to identify key challenges and data needs

Research Assistant — Applied AI in Sport 2024 – present

NSW Institute of Sport (NSWIS) · University of Technology Sydney

- Exploring the application of Large Language Models in coach and athlete environments
- Developing methods for automating content analysis to support insight generation and decision-making

Publications

Schulenkorf, N., Koparan, S., English, M., Sharp, P., Sixsmith, H., Wood, L. M., Farhart, P., & Caperchione, C. M. (2026). *Participant perceptions of HAT TRICK™ Cricket: a culturally-adapted intervention for men with South Asian backgrounds in Australia.* Health Promotion International, 41(1), February 2026. Open Access

Liu, A., Hook, A., Duffus, B., Davis, F., Hsu, H.-M., Israel, I., Donovan, J., Parker, M., Stibbe, M., Wu, M., Wilde, P., Fullinfaw, P., Uchino-Hansen, R., Koparan, S., Lind, T., Gavrielatos, T., Pestrivas, T., & Sutton, W. Eds: Mengersen, K., Wu, P., Mehta, D., Carson-Flynn, F., & Duwalage, K. I. (2026). *AusiSTAR: The NextGen Playbook 2025.* Queensland University

of Technology, March 2026. Open Access

Koparan, S. & Javadi, B. (2024). *DeepRacer on Physical Track: Parameters Exploration and Performance Evaluation*. arXiv preprint, June 2024. Open Access · GitHub

Education

Doctor of Philosophy (PhD) — Sports Data Science & AI
University of Technology Sydney 2024 – present
Next Generation Graduates Program, supported by CSIRO Data61 & Rugby Australia

Master of Information and Communications Technology — Artificial Intelligence & Cloud Computing
Western Sydney University 2021 – 2023
GPA: 6.574 / 7 · Dean's Merit Award: 2021, 2022, 2023

Bachelor of Business — Management
Western Sydney University 2017 – 2020

Conference Presentations

Generative AI for Biomechanists
Australian Institute of Sports Biomechanics Network 2026

Technical Considerations for Reliable Studies Using Large Language Models
Faculty of Health GenAI-Enhance Summit, UTS 2026

The Anatomy of a Thriving Club: Organisational Characteristics Supporting Retention and Growth
STARS, Australian Sports Commission 2025

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL
Machine Learning & AI: Scikit-learn, PyTorch, LLMs, Agents, LangChain, OpenAI API, Google Gemini API, RAG Systems
Data & Analytics: Pandas, NumPy, Statistical Modelling, Data Visualisation
Development & Tools: React, Node.js, Express, Git, REST APIs

Awards & Honours

Dean's Merit Award — Western Sydney University 2021, 2022, 2023

Supervision

Currently supervising undergraduate students on capstone research projects in sports data science and AI.